PRAHARI — Project Overview

Early Warning Health & Safety System for Hackathon

TEAM CHEAT-SHEET

1. What is PRAHARI in Simple Terms?

PRAHARI is an **early warning system** that predicts health outbreaks and workplace danger before they peak. Instead of just showing standard charts, it combines weather, air quality, hospital clinic visits, and workplace sick leave to tell health officers *what issue is rising, where, why, and what action to take right now*.

1. Weather & Air Data

Reads live heat (WBGT index) and pollution (PM2.5) levels from free open APIs (Open-Meteo & OpenAQ).

2. Health Signals

Simulates clinic visits, hospital admissions, and factory sick-leave counts per district.

3. Smart Detection Engine

Separates expected sickness (e.g., normal summer heat) from unexpected outbreaks requiring investigation.

4. Actionable Alerts

Gives clear instructions to specific roles (e.g., Site Safety Officer, District Health Officer).

2. How the System Works (4-Step Pipeline)

Step	What Happens	Why It Matters
1. Ingest	Collects weather, pollution, and health counts automatically.	Keeps data fresh without manual work.
2. Clean & Protect	Removes names and aggregates counts by location.	Guarantees 100% patient and worker privacy.
3. Analyze	ML models check for sudden unusual sickness clusters.	Prevents false alarms caused by predictable weather.
4. Alert & Direct	Sends an alert card with clear step-by-step actions.	Turns data directly into real-world decisions.

Our Winning Advantage for Judges

Most teams build simple dashboards. PRAHARI wins because it **explains why** an alert fired, **protects privacy by design**, and **proves its lead time** with built-in test benchmarks.

3. The 3 Demos We Must Show Judges

Our prototype must successfully show these three real-world health situations during the 5-minute pitch:

Scenario	Triggers & Data	Resulting Alert & Recommended Action
1. Factory Heat Stress	High WBGT Heat + Rising Sickness Leave at 2 outdoor sites.	**Alert:** Worksite Heat Hazard. **Action:** Set up shade stations & hydration breaks.
2. City Pollution Surge	High PM2.5 Air Pollution + Rising Asthma/Cough visits.	**Alert:** Respiratory Threat. **Action:** Issue public advisory & distribute masks.
3. Unknown Outbreak	Rising Stomach/Fever cases with *normal* weather data.	**Alert:** Unexplained Cluster. **Action:** Send Rapid Response Team to test local food/water.
4. Missing Data Test	Turn off clinical feed manually during demo.	**Alert:** System lowers confidence badge without crashing or firing false alarms.

4. Team Roles & Task Split

Member / Role	Core Tasks	Main Output
Data & Backend	Set up Open-Meteo API ingestion & PostgreSQL database.	Daily data feed & clean database tables.
Machine Learning	Build synthetic data generator & LightGBM anomaly detection model.	Alert score & SHAP explainability script.
Frontend UI	Build React map dashboard, alert inbox, and evidence card.	Interactive Web App hosted on Azure.
Governance & Demo	Write YAML action playbooks, prepare slide deck & presentation.	Winning pitch deck & validation report.

5. Critical Rules to Avoid Mistakes

- **No Real Patient Data:** We will generate realistic fake health data (`simulator/`) linked to real historical weather.
- **Zero Cloud Cost Spikes:** Train ML models locally on laptops or Colab. Do not run heavy compute engines on Azure.
- **LLM is Optional:** The core app must work with plain template text. We only use AI rephrasing if API quotas allow.
- **Always Show Privacy Badges:** Display cell-suppression and aggregate notices on every screen.